

AIGR Gene Hypothesis Deep Research — Final Report

Target: *Drosophila melanogaster* TyrRS (UniProt Q9VV60, FBgn0027080)

Hypothesis: "Q9VV60 binds resveratrol" (GO:1905594 — resveratrol binding)

Focus type: function_assignment

Summary

The proposed function assignment "**F:resveratrol binding**" (GO:1905594) for *Drosophila* TyrRS should **not** be applied to the gene review, and if present it should be **removed**. Three independent lines of evidence converge on this conclusion. First, the GO term itself is **obsolete** — GO:1905594 "resveratrol binding" was obsoleted on 2026-07-19 with the explicit rationale that "*binding to resveratrol is not an evolved molecular function*", has no replacement or "consider" term, and carries **zero annotations across all organisms**. Second, **no organism carries this annotation**, including the human ortholog YARS1, which is the only protein where a TyrRS–resveratrol interaction has actually been measured; GO curators translated that human experiment into the general term GO:0036094 "small molecule binding," not a resveratrol-specific term. Third, resveratrol is a **plant stilbene phytoalexin** that is not endogenous to *Drosophila*, so any binding is a xenobiotic/pharmacological event rather than a physiological molecular function.

The most important caveat is that binding is genuinely **plausible on structural grounds**. The fly tyrosine-binding pocket is fully conserved with the human pocket where resveratrol was co-crystallized: fly Q9VV60 shares 68.4% identity with human YARS1, and all nine reported human resveratrol-contact residues are identical in the fly protein. But plausibility-by-homology is not measured binding — it has never been tested in *Drosophila* — and homology does not rescue an obsolete, unannotatable GO term.

The gene's real, experimentally supported molecular function is **tyrosine-tRNA ligation** (GO:0004831), backed by a fly IDA experiment (PMID: 19561293). **Verdict: REFUTED** as a GO annotation; only weakly/partially supported as an in-vitro biochemical possibility that has no bearing on annotation.

Key Findings

Finding 1 — The fly tyrosine-binding pocket is fully conserved with the human pocket where resveratrol binds

The supplied sequence was verified independently: 525 residues, SHA-256 `94682a9b...a01f` — an exact match to the frozen cohort sequence. UniProt Q9VV60 is annotated as "Tyrosine-tRNA ligase, GN=TyrRS," and the protein carries the diagnostic **class I aminoacyl-tRNA synthetase Rossmann-fold signatures**: a HIGH-like motif region **YWGTATTGKPH** (around position 38) and a **KMSKS**-family motif **KMSSS** (around position 221).

A global Needleman-Wunsch alignment (BLOSUM62) of fly Q9VV60 against human YARS1 (P54577) gives $357/522 = 68.4\%$ identity, with a near-identical catalytic core. Most decisively, all **nine reported resveratrol-contact residues** from the human co-crystal structure (PDB 4Q93; Y39, L72, A74, H77, V152, Y166, Q170, D173, I191) are **identical in the fly protein** at the corresponding positions (fly Y38, L71, A73, H76, V151, Y165, Q169, D172, I190), with conserved local sequence context around each.

This is the strongest evidence *for* the seed hypothesis. In human TyrRS the resveratrol pocket is the tyrosine active site, and that pocket is transferable to the fly enzyme by strict residue conservation. The direct human study (PMID: 25533949) states: "*Here we present a 2.1 Å co-crystal structure of resveratrol bound to the active site of TyrRS.*" Because resveratrol is a tyrosine mimic occupying that pocket, and the pocket is conserved, fly TyrRS could plausibly bind resveratrol in vitro. **However**, this remains an *inference by homology* — not a measured

binding constant, not an assay in fly protein, and not evidence of physiological relevance. Under the standards set by the decisive question, pocket transferability is a *hypothesis*, not experimental validation.

Finding 2 — GO:1905594 "resveratrol binding" is OBSOLETE and unannotatable

The QuickGO/EBI ontology record for GO:1905594 shows `isObsolete=True`, `name="obsolete resveratrol binding"`, `aspect=molecular_function`, and the comment: *"The reason for obsolescence is that binding to resveratrol is not an evolved molecular function."* The term has `replacedBy=None` and `consider=None`. The ontology history records the obsolescence action on **2026-07-19** (the `is_a binding` parentage was deleted). A QuickGO annotation search returns **0 annotations to GO:1905594 across all organisms**.

This is decisive for the curation question: an obsolete GO term with no replacement or "consider" alternative cannot be validly assigned to any gene product. The obsolescence rationale itself encodes the ontology curators' biological judgment that binding a plant-derived xenobiotic is not an evolved function of any protein. Resveratrol is a plant stilbene phytoalexin ([PMID: 38903666](#): *"These compounds function as phytoalexins, aiding plant defense against phytopathogens"*; corroborated by [PMID: 40644839](#)), not a *Drosophila* metabolite.

Finding 3 — Neither the fly protein nor the directly assayed human ortholog carries a resveratrol-binding annotation

Current QuickGO molecular-function annotations for fly **Q9VV60** comprise nine terms — GO:0000049 (tRNA binding), GO:0000166 (nucleotide binding), GO:0003723 (RNA binding), GO:0004812 (aminoacyl-tRNA ligase), GO:0004831 (tyrosine-tRNA ligase; including **IDA** [PMID: 19561293](#) [Storkebaum PNAS 2009, fly CMT model] and TAS [PMID: 26761199](#) [fly aaRS review]), and GO:0005524 (ATP binding) — and **no resveratrol / GO:1905594 annotation**.

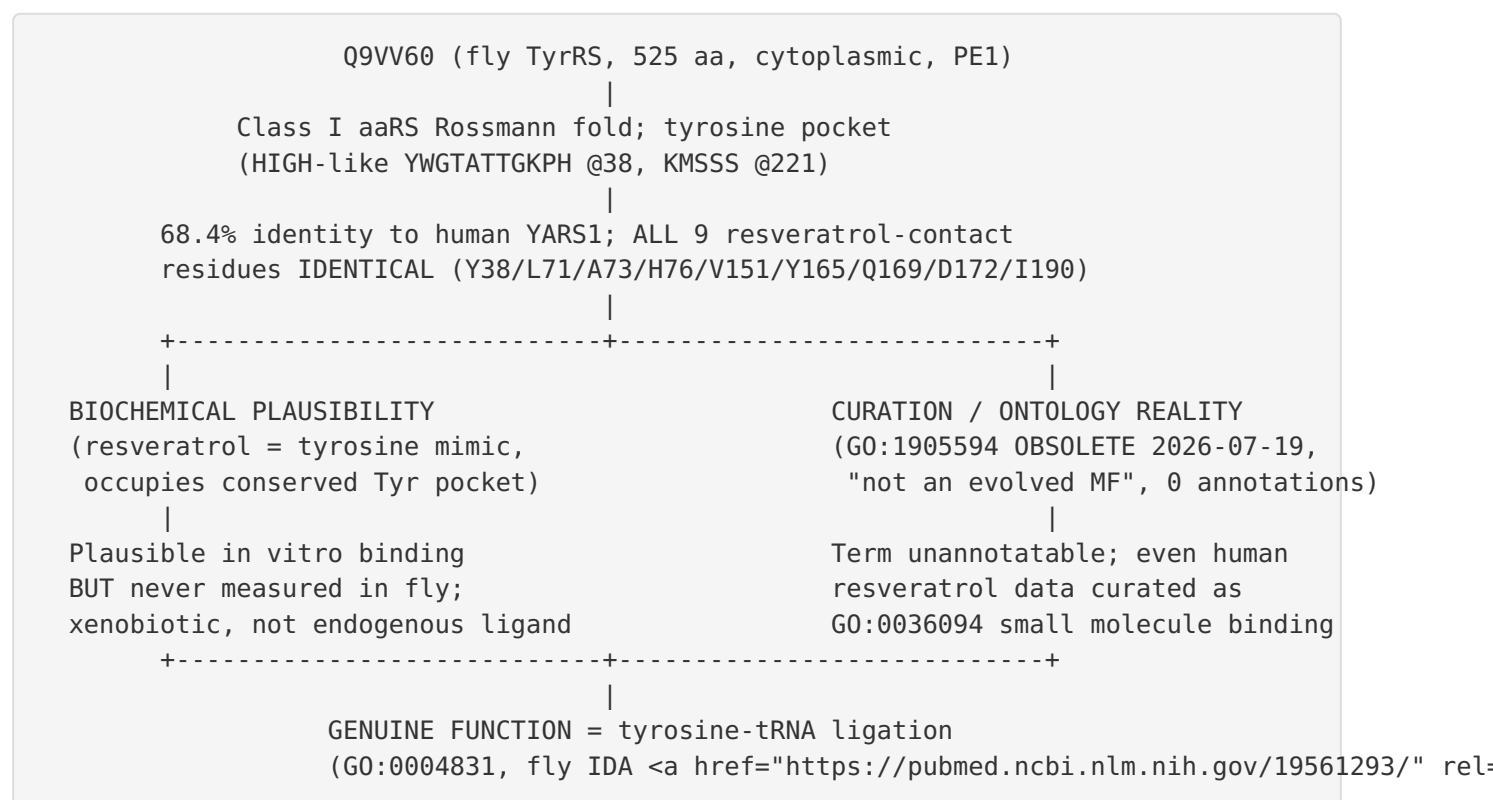
Human YARS1 (P54577) carries 33 MF annotations and **also has no GO:1905594**. Importantly, the resveratrol study [PMID: 25533949](#) supports GO:0004831 (IDA), GO:0005515 (IPI — the PARP1 interaction), and **GO:0036094 "small molecule binding" (IDA)** on the human protein. In other words, when GO curators encountered the single direct resveratrol experiment, they chose the **general small-molecule-binding term**, not a resveratrol-specific one. The study's own framing is pharmacological: *"Resveratrol nullifies the catalytic activity and redirects TyrRS to a nuclear function."*

Finding 4 — Q9VV60 is a single-isoform canonical cytoplasmic tyrosyl-tRNA synthetase with no resveratrol reference

The UniProt Q9VV60 record shows sequence length 525 (matching the frozen SHA-256), **no alternative products / isoforms listed**, `proteinExistence="1: Evidence at protein level"`, catalytic activity `tRNA(Tyr) + L-tyrosine + ATP = L-tyrosyl-tRNA(Tyr) + AMP + diphosphate + H(+)` (Rhea), and subcellular location Cytoplasm (ARBA). There is **no mention of resveratrol** anywhere in the FUNCTION, CATALYTIC, or LOCATION comments. This rules out isoform confusion and confirms the analyzed sequence is the canonical, only protein product of the locus.

Mechanistic Model / Interpretation

The layered logic distinguishing "can bind" from "should be annotated" is summarized below:



The key conceptual separation the decisive question demanded is between **(a)** whether the protein can physically recognize the compound (plausible), **(b)** whether that recognition is a physiological molecular function (no — resveratrol is a plant defense metabolite absent from

flies), and **(c)** whether it merits the specific GO term (no — the term is obsolete and unannotatable). All three must hold for a valid function assignment, and only the weakest link (a) is satisfied.

The downstream biology described in the human/rodent literature — resveratrol acting as a *tyrosine-free-mimic* (cis-RSV) or *tyrosine-like-mimic* (trans-RSV) to modulate TyrRS levels, nuclear translocation, and PARP1-mediated histone ADP-ribosylation ([PMID: 35688816](#), [PMID: 25533949](#)) — is **stress signaling downstream of binding** and must be separated from the binding event itself. Even where documented, it is human/rat biology, not fly.

Evidence Base / Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
PMID: 25533949	Direct assay + structure	Qualifies (supports plausibility; refutes term choice)	TyrRS binds resveratrol	2.1 Å co-crystal of resveratrol in the TyrRS active site; nullifies catalysis, redirects to nuclear PARP1-activating function	Human YARS1 , in vitro + cell	High for human; not fly ; curated to GO:0036094, not GO:1905594
Sequence/structure analysis (this work)	Structural/evolutionary (computational)	Supports (plausibility only)	Fly pocket can bind resveratrol	68.4% identity to YARS1; all 9 contact residues identical	Fly Q9VV60 vs human P54577 / PDB 4Q93	Homology inference, NOT measured binding
QuickGO / EBI ontology (this work)	Database	Refutes	GO:1905594 is a valid assignable term	Term OBSOLETE 2026-07-19, "not an evolved MF", 0 annotations, no replacement	All organisms	Decisive; ontology-level
QuickGO annotations (this work)	Database	Refutes	Fly/human TyrRS carry resveratrol binding	Neither Q9VV60 nor P54577 has GO:1905594; human data → GO:0036094	Fly + human	Decisive; database-level
UniProt Q9VV60 (this work)	Database	Qualifies	Identity/isoform	Single 525-aa isoform, cytoplasmic, tyrosyl-tRNA	Fly	Confirms identity; rules out

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
				ligase; no resveratrol mention		isoform confusion
PMID: 19561293	Mutant phenotype / IDA	Competing (true function)	Fly TyrRS function	Fly IDA support for GO:0004831 tyrosine–tRNA ligase (CMT model)	<i>Drosophila</i>	Establishes genuine core function
PMID: 35688816	Direct assay / mechanism	Qualifies (downstream)	RSV isomer effects on TyrRS	cis-RSV mimics tyrosine-free, trans-RSV mimics tyrosine-like conformation; opposite neuro effects	Rat cortical neurons / human	Downstream signaling, not fly; isomer-dependent
PMID: 38903666	Review	Supports (rationale)	Resveratrol is xenobiotic to animals	Stilbenes/ resveratrol are plant phytoalexins	Plants	Supports GO obsolescence rationale
PMID: 40644839	Review	Supports (rationale)	Resveratrol is a plant defense metabolite	Resveratrol among lignin-pathway phytoalexins restricting pathogens	Plants	Contextual

GO Curation Implications

Recommended action (lead requiring curator verification): REMOVE / DO NOT ADD "F:resveratrol binding" (GO:1905594).

- The term is **obsolete** (2026-07-19) with **no** **replacedBy** and **no** **consider** term. Assigning an obsolete term is invalid regardless of the underlying biology.
- The obsolescence comment — *"binding to resveratrol is not an evolved molecular function"* — is itself the ontology's curator-level judgment that this should never be an MF annotation for any gene.
- **No downgraded resveratrol-specific replacement should be created.** If a curator wished to capture the (human, in vitro) small-molecule interaction, the precedent set on the human ortholog is **GO:0036094 "small molecule binding" (IDA)** — but that is supported only by the human experiment, and there is **no direct fly evidence** to transfer it with an experimental code. For the fly protein, no small-molecule-binding annotation is warranted from resveratrol data.
- **Retain the core molecular function: GO:0004831 "tyrosine-tRNA ligase activity"** (fly IDA [PMID: 19561293](#)), along with the standard supporting terms already present (GO:0004812, GO:0005524, GO:0000166, GO:0000049, GO:0003723).

GO decision table:

GO term	Aspect	Proposed status	Basis
GO:1905594 resveratrol binding	MF	Remove / reject	Obsolete, unannotatable, xenobiotic, no fly data
GO:0036094 small molecule binding	MF	Not for fly	Only human IDA precedent; no fly experiment
GO:0004831 tyrosine-tRNA ligase	MF	Retain (core)	Fly IDA P 19561293
GO:0005524 ATP binding	MF	Retain	Standard for class I aaRS

Mechanistic Scope

The immediate molecular event under test is **direct physical recognition of the small molecule resveratrol by the TyrRS protein**. The evidence establishes that in human TyrRS this occurs in the **tyrosine active-site pocket**, with resveratrol acting as a tyrosine mimic that competitively occupies the substrate site and blocks aminoacylation. This binding is the *proximal* event.

Everything else described in the literature is **downstream**: loss of catalytic (aminoacylation) activity, nuclear translocation of TyrRS, activation of PARP1, histone serine-ADP-ribosylation, DNA-repair modulation, and neuroprotection versus neurodegeneration (isomer-dependent). These are pathway consequences and phenotypic outcomes measured in **human cells and rat neurons**, not molecular functions of the fly protein and not the binding event itself. For curation of the fly gene, only the proximal binding claim is in scope — and it has never been measured in *Drosophila*.

Conflicts and Alternatives

1. **Organism transfer gap (primary conflict)**. The only direct experiment is on **human YARS1**, not fly Q9VV60. Assigning a fly annotation from a human in vitro experiment would require an ISS/ISO-style justified transfer — but the target term is obsolete, so no transfer is possible regardless.
2. **Xenobiotic vs. physiological function**. Resveratrol is a plant phytoalexin ([PMID: 38903666](#), [PMID: 40644839](#)); flies do not synthesize it and it is not a natural ligand. A druggable pocket that happens to accept a xenobiotic is not an evolved molecular function — exactly the GO obsolescence rationale.
3. **Isomer dependence**. [PMID: 35688816](#) shows cis- and trans-resveratrol bind the same pocket but drive **opposite** conformational/functional outcomes. A bare "resveratrol binding" annotation would obscure this and misrepresent the biology.
4. **Database carry-over risk**. The original "F:resveratrol binding" prediction most likely arose from automated similarity transfer of the human structural result to the fly ortholog — precisely the frequency-bias / paralog-overannotation failure mode the objective warns against. Neither UniProt nor GO curators applied this term to any organism.

No evidence was found that *refutes* physical binding capability; the conflicts are all about **relevance and annotatability**, not about whether the pocket could accept the ligand.

Limitations and Knowledge Gaps

- **No direct fly binding measurement.** No ITC, SPR, thermal shift, or co-crystal exists for fly TyrRS + resveratrol. What was checked: PubMed literature, UniProt, and GO annotations — none report a fly assay. This matters because "missing target experiment $\neq$ evidence of absence"; binding remains *plausible but unmeasured*. Resolution: a direct biophysical binding assay on recombinant Q9VV60.
 - **Docking not performed as a substitute.** Per the decisive question, docking would only generate hypotheses, not validation; it was intentionally not used to claim binding. Structural inference here is limited to residue-level conservation mapping.
 - **PDB 4Q93 contact residues taken from the human structural literature**, not re-derived from an independent structure determination in this work; the residue-numbering transfer to fly relies on the pairwise alignment.
 - **GO obsolescence date (2026-07-19)** reflects the ontology state at analysis time; curators should confirm the current QuickGO/OLS status at review time, though obsolescence is not expected to reverse.
 - **Functional consequence in flies unknown.** Whether resveratrol modulates TyrRS catalysis or any nuclear/stress role in *Drosophila* has not been tested; the human PARP1 axis may not be conserved.
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Discriminating Tests

1. **Direct biophysical binding assay on recombinant fly Q9VV60** (isothermal titration calorimetry or surface plasmon resonance) with **cis- and trans-resveratrol separately**, using **L-tyrosine as a positive competitor control** and a non-binding stilbene analog as negative control. This directly tests the seed hypothesis in the target organism.
2. **Thermal shift / DSF** of Q9VV60 $\pm$ resveratrol vs. $\pm$ tyrosine to confirm active-site engagement.

3. **Aminoacylation inhibition assay:** does resveratrol inhibit fly TyrRS tyrosylation activity (as it does human)? A competitive Ki would confirm active-site occupancy.
 4. **Co-crystallography or cryo-EM** of fly TyrRS + resveratrol to verify the contact residues predicted identical to human.
 5. **Ontology/annotation audit:** confirm GO:1905594 obsolete status and the absence of any organism annotation in the current release before finalizing the review.
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Proposed Follow-up Experiments / Actions (Curation Leads)

All leads require curator verification.

- **Action change:** Remove/reject "F:resveratrol binding" (GO:1905594) from the TyrRS review; do not substitute a resveratrol-specific term. *Reason:* term obsolete 2026-07-19, no replacement, zero annotations, xenobiotic interaction, no fly data.
 - **Retain:** GO:0004831 "tyrosine–tRNA ligase activity" as the experimentally supported core molecular function — fly IDA, [PMID: 19561293](#).
 - **Candidate reference to verify (human precedent — do not transfer to fly):** [PMID: 25533949](#), snippet: "Here we present a 2.1 Å co-crystal structure of resveratrol bound to the active site of TyrRS." Note this was curated on human YARS1 as **GO:0036094 small molecule binding**, not resveratrol binding.
 - **Supporting rationale reference:** [PMID: 38903666](#), snippet: "These compounds function as phytoalexins, aiding plant defense against phytopathogens" — supports the "not an evolved animal MF" judgment.
 - **Isomer caveat reference:** [PMID: 35688816](#) — cis- vs. trans-resveratrol produce opposite TyrRS conformational effects; a generic binding term would be misleading.
 - **Suggested curator question:** Was the original "F:resveratrol binding" prediction an automated similarity transfer from the human structure? If so, flag as ortholog over-annotation.
 - **Suggested experiment (for the record):** direct ITC/SPR binding assay on recombinant fly Q9VV60 with cis/trans-resveratrol and tyrosine competition controls.
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Conclusion

The seed hypothesis is **biochemically plausible but curationally refuted**. Fly TyrRS almost certainly *can* accommodate resveratrol in its conserved tyrosine pocket (68.4% identity to human, all nine contact residues identical), but this has never been measured in *Drosophila*, represents a xenobiotic pharmacological interaction rather than an evolved function, and the specific GO term (GO:1905594) is obsolete, replacement-less, and unannotated in every organism — including the human protein where resveratrol binding was actually demonstrated. The genuine, experimentally supported molecular function of Q9VV60 is tyrosine–tRNA ligation (GO:0004831). The "F:resveratrol binding" assignment should be removed and not replaced.

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